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The Palm OS® Cobalt details: Core OS, Multithreading, etc.

The new foundation

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Core OS changes

Lowest levels of the system changed dramatically

- Per-process memory protection for stability & security
- Heavy use of multiple processes and threads
- Consistent frameworks to extend architecture
  - IPC framework
  - Graphic subsystem framework
  - I/O framework and driver model
  - STREAMS for higher-level protocols
  - Connection Mgr “framework”



Definitions

Threads are a unit of execution

- They have stacks, registers, priority, some unique data

Processes are execution environments

- They have permissions to heaps (r/o, r/w or none)
- Memory access privileges are enforced by the MMU & Kernel
- Processes are given “keys” which grant access or permissions to services (e.g. memory, servers…)



Palm OS® Cobalt Processes and Memory Mapping

Data Mgr	System	I/O	Application	Background	Sublaunch x3	Licensee
Data Mgr threads	Window Mgr	IOS	UI application	Comms thread(s)	Notification Agent	License UI Component
	App Mgr	VFS	PACE	Exchange Sublaunch	Find Agent	Licensee Srv
	Event Mgr	STREAMS framework	OS Shared libraries	Slips	PACE	
	AM/AZM	STREAMS components		Sound callbacks		
	Status Bar	HW device drivers		PACE		
	Feature Srv					
	Exchange Srv					
	Notification Srv					
Local Heap	**Local Heap**	**Local Heap**	**Local Heap**	**Local Heap**	**Local Heap**	**Local Heap**

System Shared Memory

Secure DBs | **Frame Buffer** | **Audio Mixer** | **Legacy Heap (on request)**

Storage Heap

ROM



Threading APIs

Yes, there are threading APIs!

- SysThreadCreate, SysThreadStart: in your (app) process
- SysAppLaunchRemote and SysAppLaunchBackground
- UI in other threads/processes
- Send events to another thread's event queue
- Atomic operations: Add32, And32, Or32, CompareAndSwap32
- Simple synchronization primitives
- Thread-specific data



Security

When a UI app starts, it gets its own process

- Newly created process, default permissions
- Given a fresh dynamic heap, plus other heaps
- Can specify security for the libraries it loads
- Can have sublaunching happen in a remote process
  - E.g. Find, Notifications…



New ARM native runtime model

All executable modules have the same structure

- (e.g. applications, shared libraries and plug-ins)
- Multiple modules in one PRC, and multiple entry points for each

Each loaded module, in every process, has its own globals

Each gets launch codes (Reset, sysNotify, Initialize, Finalize…)

Shared libraries are executed as subroutines of the app, with its rights

System APIs are shared libraries

- Many simply do IPC to a server process



Patches

Program Loader supports patching

A Patch has normal code and data segments, plus a resource of type ‘amdp’ (ARM Module Patch) to tell what it patches

Patches and the patched library are always loaded and unloaded together

- Can also call SysRegisterPatch if not in a prc with type ‘apch’

Multiple patches can be on same entrypoint



Patches

Only trusted patches are loaded in secure processes, via code signing

Much system code runs in secure processes

- Therefore can't be patched like before

App and/or licensee can prevent being patched



Memory Manager, Heaps, etc.

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New Memory Mgr and environment

Single 32-bit virtual address space, divided into segments

- Segments are basis for memory protection
- Heaps are allocated in segments

Heaps are bigger and smarter

- Heaps can be up to 256MB
- Individual allocations up to 16MB
  - (Assuming there's RAM of course)
- These limits are internal; API wouldn't change to increase
- RAM isn't committed till a heap needs it

Data Mgr is now separate;

- calling MemMgr on storage heap just calls through to the the Dm API.



Sharing memory

Legacy heap

- MemPtrSetOwner(0) was a hack, and goes against the “create a clean and fresh process” philosophy
- Chaining to another application and passing it data (in cmdPBP) still works, even for linked lists, because PACE uses the “legacy heap” for 68K apps.
- Legacy heap can be mapped into a native app’s process upon request.

Feature Pointers, Databases are better choices for sharing data



Database access

The Data Mgr manages two segments

First is traditional storage heap

- Directly readable by apps

Second segment is for protected or secure databases

- The Data Mgr will give you copies of the data
- Communication to other process happens automatically



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Higher layers



Application Programming Model

Prior to Palm OS® Cobalt, the app was in full control

- The single event loop was controlled by the app
- Had direct access to everything
- Stability and user experience depended on app behaving properly

That model was limiting

- Simplicity of development, but simple apps and system
- System services weren't scalable
- Apps cannot easily co-exist and interoperate



Application Programming Model

Palm OS Cobalt model: enforce security and reliability

- PACE app still thinks it “runs the show” but the system is in charge
- New app can use new services
- The application sees what it is supposed to

Standard application-level threading

- Eases communications code
- Enables background tasks

More dynamic and responsive UI

Like many modern desktop systems, but tuned



Events

Designed around Multiple Event Queues

Each is paired with a process or thread and is managed by the Window Mgr.

Some former events now won't be seen by applications

- Some are now notifications
- Some are delivered to other processes than the app
- (There are now better mechanisms available than sending events all the time)



Programming tidbits

Standard data types (stdint.h style)

- uint8_t, int32_t and so forth

libc functions and STLPort

C++ ABI still in flux so no exposed C++ functions

- (yet!)

VFS and localization: no substantial change

- A few improvements



Data Manager

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Three Types of databases

Classic: same as 5.x and prior

- Records and resources

Extended: like Classic, but better

- Supports >64k records or resources
- Uniquely identified by name plus creator ID
- Likely to have data in little-endian format

Schema

- The new kind! Flexible tables of data.



Compatibility

Old databases (restored via HotSync®) are Classic

- Situation for apps in PACE doesn't change
- Native apps can use old APIs (some now v50)
- Data isn't changed for endianness/alignment by the system

Extended and Schema databases can't be loaded on a pre-Palm OS® Cobalt device

- Classic can, via ExgDbWrite



PIM databases

The PIM applications use schema databases

- Schema definitions will be documented
- Old databases might not exist, or might be ignored
- (We've warned about this for years)

PIMs also support various launch codes, Exchange Manager, etc.

- Simple importing can still be simple (and OK from PACE)



Secure Databases

The Data Mgr can control access to databases

- Uses Authorization Manager to do so

DbCreateSecureDatabase

- When new, is completely protected: not even you can open it!

You then grant Read, Write, Delete, SchemaEdit, Backup, and Restore privileges

- With the Authorization Mgr and Authentication Mgr APIs
- Rules can require a digital signature, password, PIN, or unrestricted access



Secure Databases (cont)

Security is per database, not per row

Secure dbs will be visible with DmFindDatabase, but not necessarily openable

To allow sync, must give access to HotSync®

- Can qualify for Read, Write, Delete, SchemaEdit, Backup, Restore.
- Only to trusted desktops
- Backup of a secure database will be encrypted on the desktop

If not given access, no backup is possible!



General Data Manager Topics

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Resource search chains

Resource chains are not good choices any more

- In Palm OS® Cobalt, there are a lot more shared libraries that might need their own resource databases open
- Having a resource chain is unhelpful when using >1 threads

Therefore, the resource chain is deprecated

- Only supported for compatibility reasons, with Palm OS® Garnet versions of existing database open calls

Many Resource-focused APIs now need a DmOpenRef (e.g. FrmGotoForm)

Also, Card parameters have been removed



Database Name Scope

Prior to Palm OS® Cobalt, database names must be globally unique

In Palm OS® Cobalt, namespace is name + Creator ID.

3 namespaces: for classic, extended, and schema

New APIs specify the namespaces to search

- Can search multiple at once

For APIs that return a single database (e.g. DmFindDatabase), the first match is returned



No Mixing with the Memory Mgr

Data Manager now manages its own memory

- DmHandleLock
- DmHandleResize
- DmHandleSize
- DmHandleUnlock
- DmHandleFree
- DmPtrResize
- DmPtrSize
- DmPtrUnlock
- DmRecoverHandle



Compatibility and interoperability

PACE apps use Classic databases

Native apps can use Classic, Extended or Schema.

- Might have to worry about endianness, alignment.

Good user experience is crucial

- Consider if/when you change endianness
- Consider when to use Extended databases
- Use "wire format" for beaming etc.

Conduit implications



Questions and Answers?

Come to the labs tonight for more info and help

Protein-forum@news.palmos.com next week!



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Contact: Doug MacMillan
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Contest: Open Labs (Tues/Wed)
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